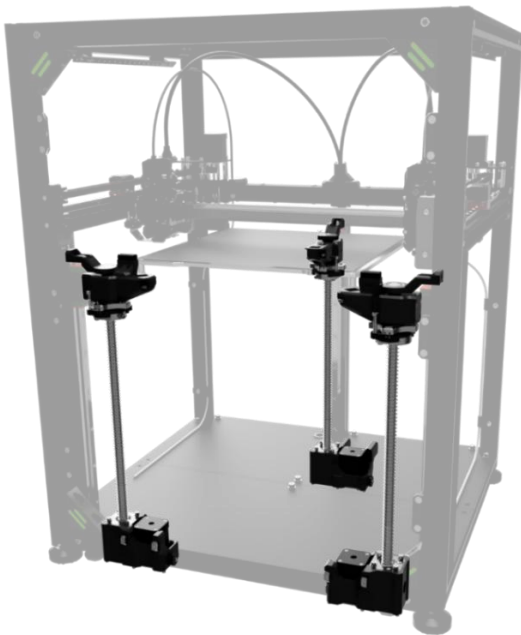


	Z-Upgrade 4.0 for VCORE 4.0/4.1	BRS-Engineering

Z-Upgrade 4.0 manual

Evolutions

Indice	Date	Description de l'évolution	Auteur
0.0	30/01/2026	Création	FBR
0.1	25/03/2026	1204 implementation	FBR
0.2	28/03/2026	1204 FK10 simplification	FBR
0.3	25/04/2026	Locknut deletion + Troubleshooting implementation	FBR
0.4	19/08/2026	Heavy duty (HD) addition	FBR



1 Introduction

First, a big thanks if you have purchased the kit from my store, it helps me to continue development and fuel my passion

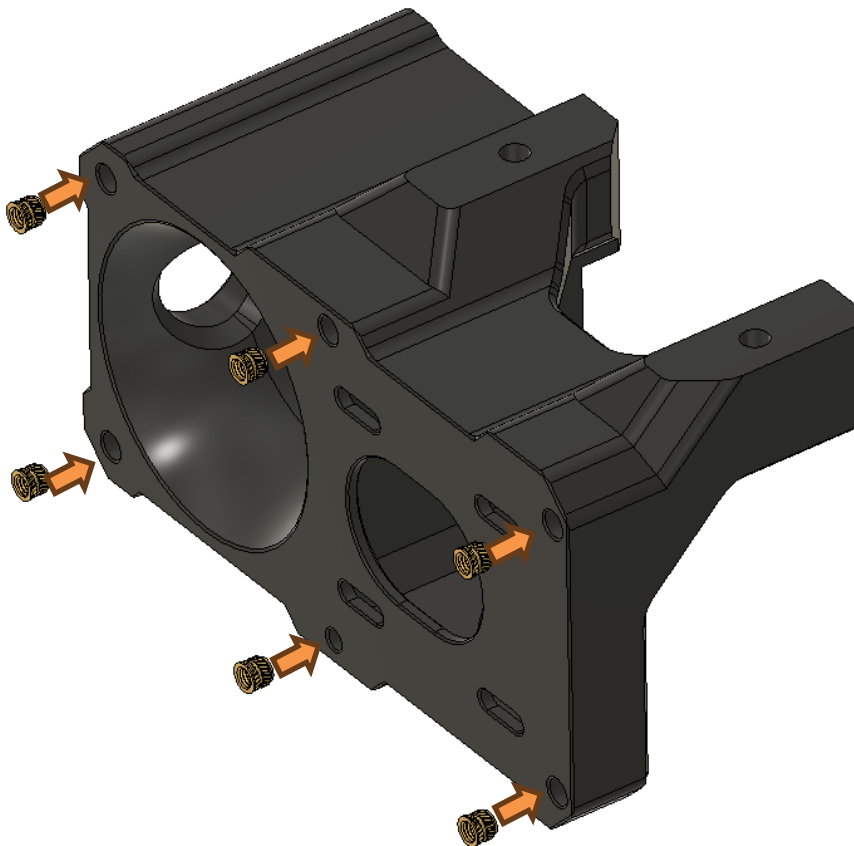
If you are a Hobbyist and make this mod on your side as DIY, welcome to the project!

2 Parts preparation

Those steps don't concern you if you bought your kit at BRS store, as I deliver it pre-assembled

Mainparts commons

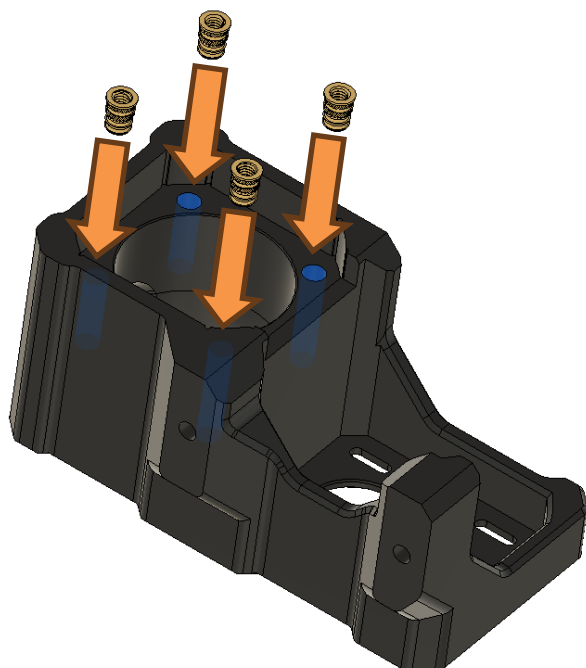
Place 18x M3 inserts over the main blocks, All 3 blocs have the same layout, 6 on each.



FOR LEGACY / 1204 VERSION:

Insert the M4x10mm insert in the top of the part

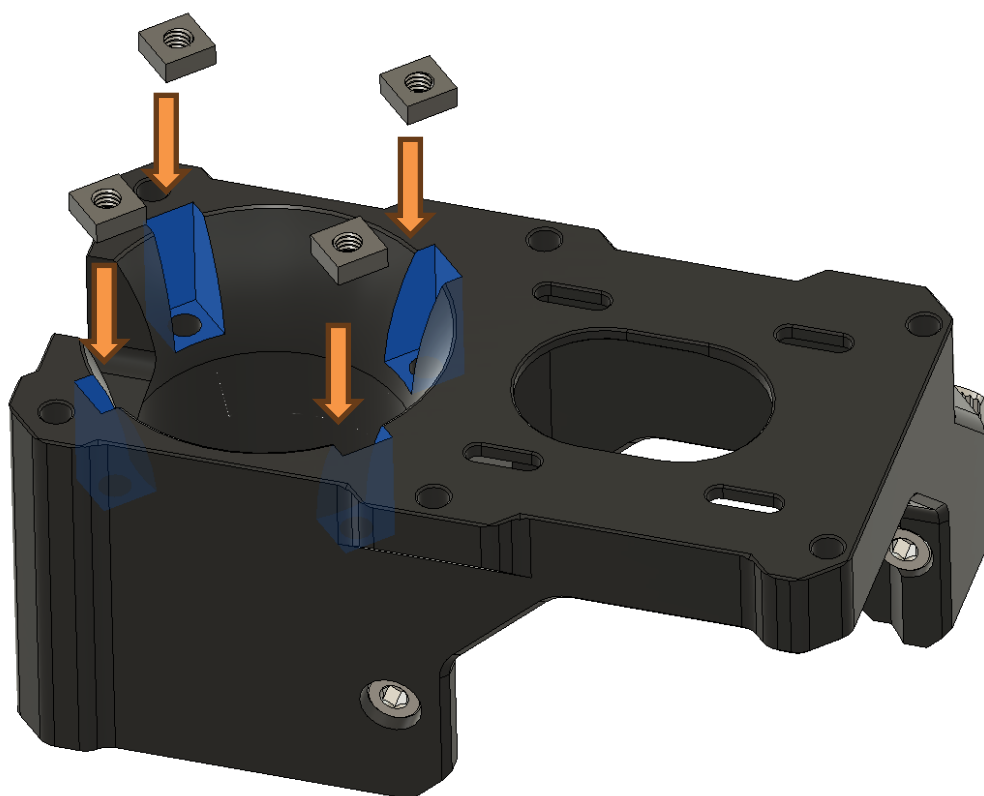
A long M4 insert here is needed, the goal is to provide a strong bond

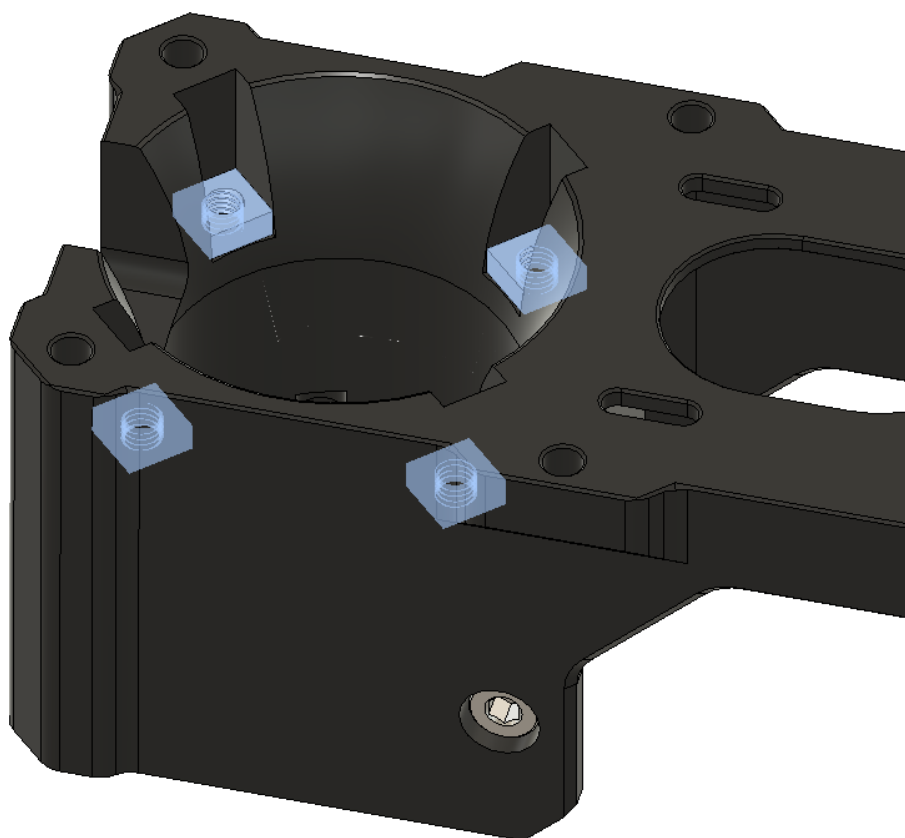


FOR HEAVY DUTY (HD) 1605 Only

Insert the M4 nut in the openings bellow

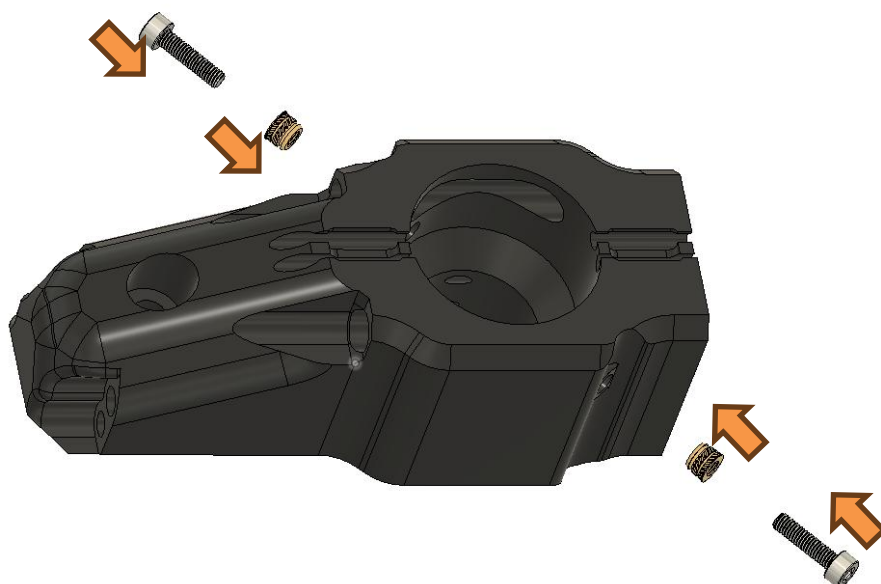
Pressfit them against the square surface

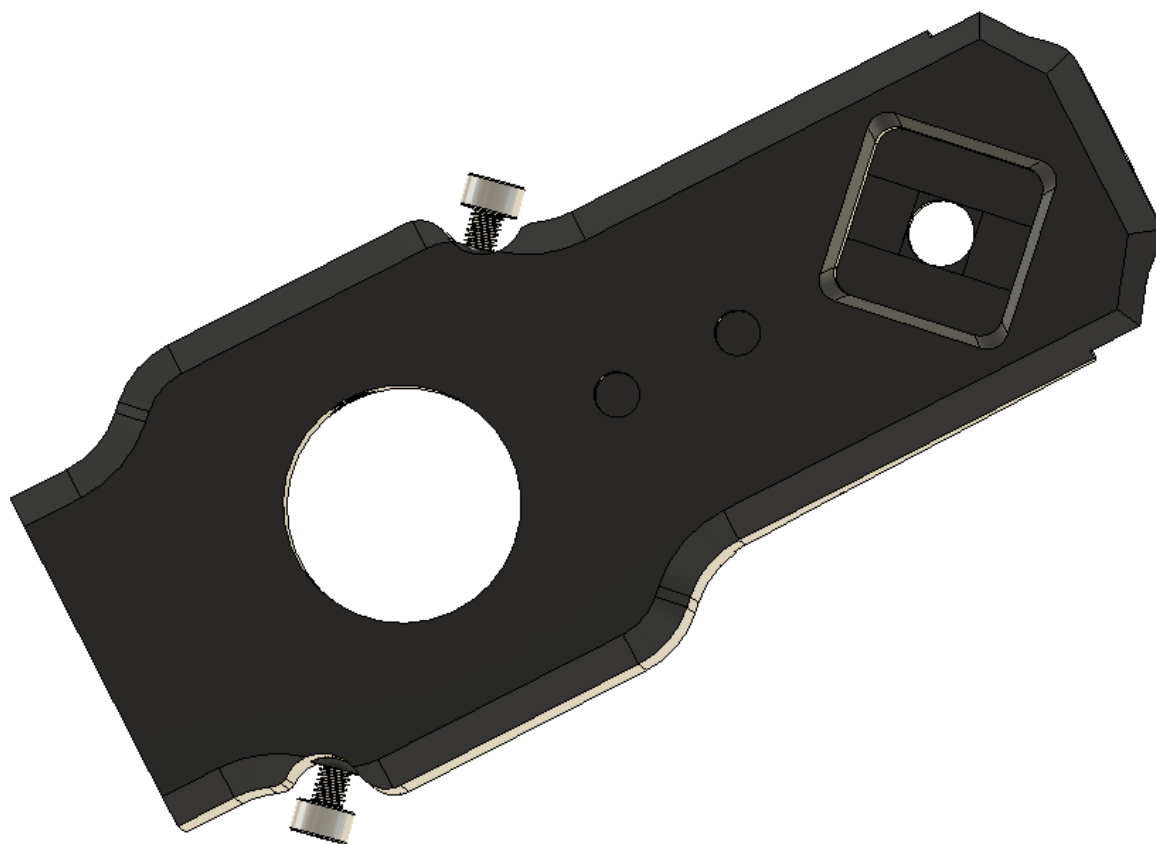




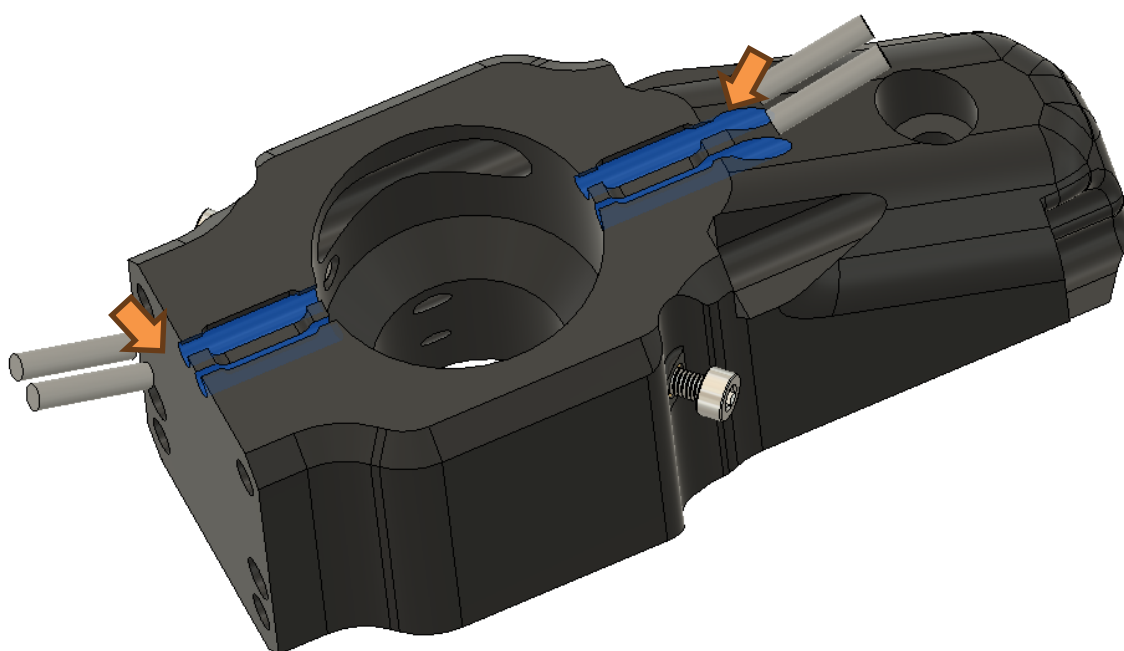
Arms

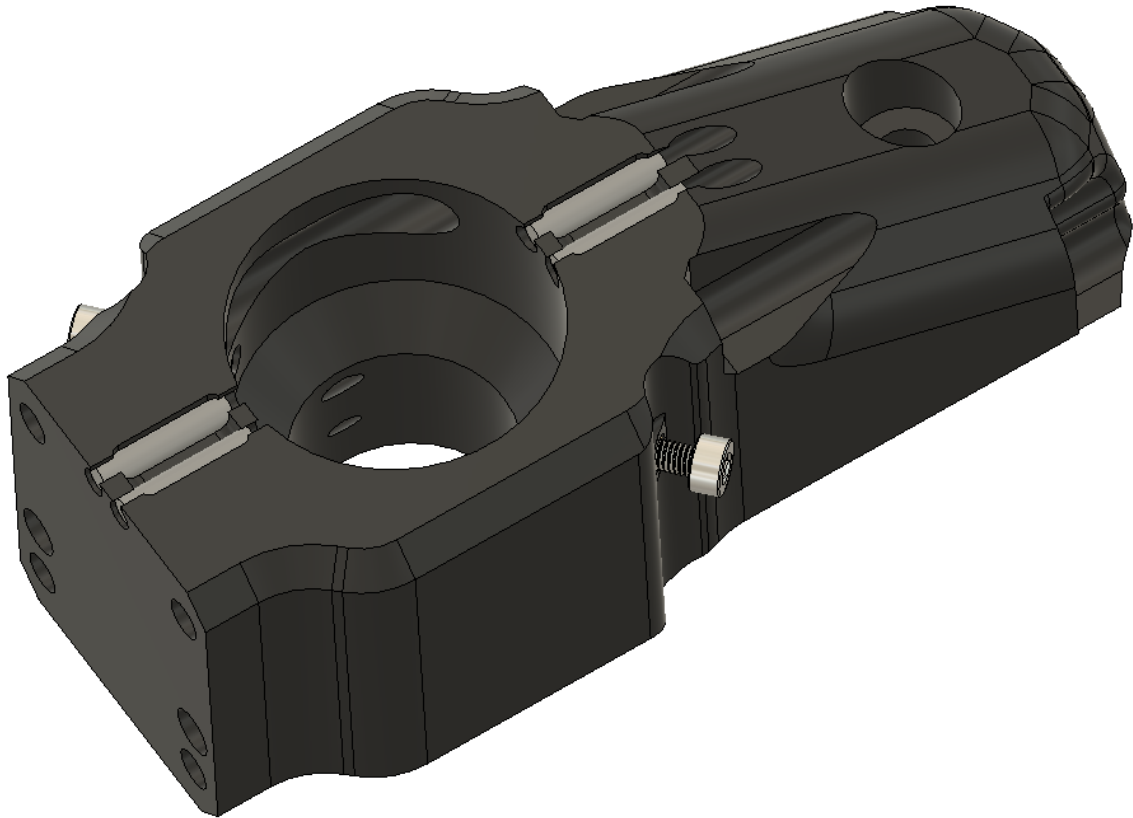
On each arms, insert 2x M3 inserts followed by the 2 M3x12mm



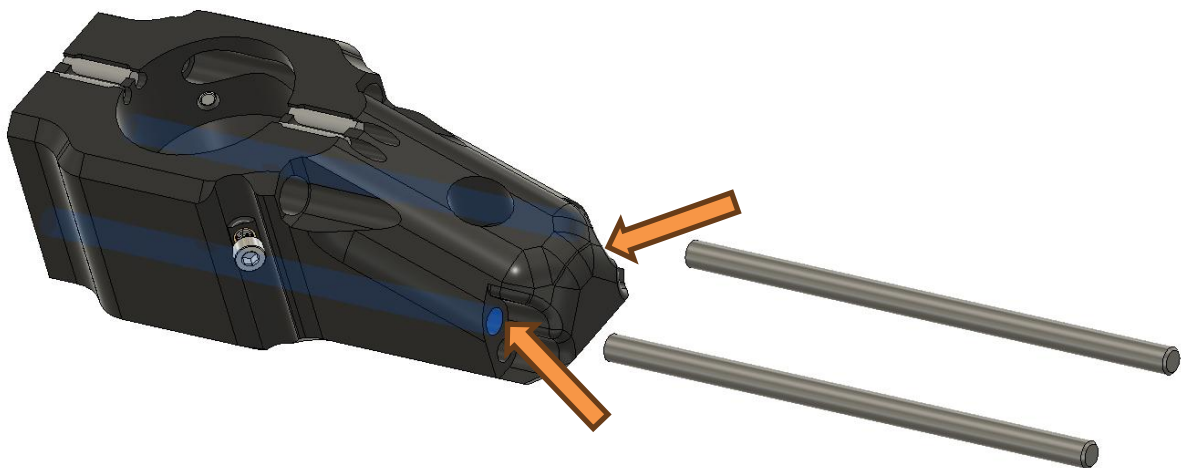


Place the 3mm pins provided with WobbleX units (***BRS kit purchased features those pins already in place***)





If you want to reinforce the arms to use a custom bed, larger or heavier and want to be future proof, the 2 holes in each arms are here to accept 5x100mm steel rods



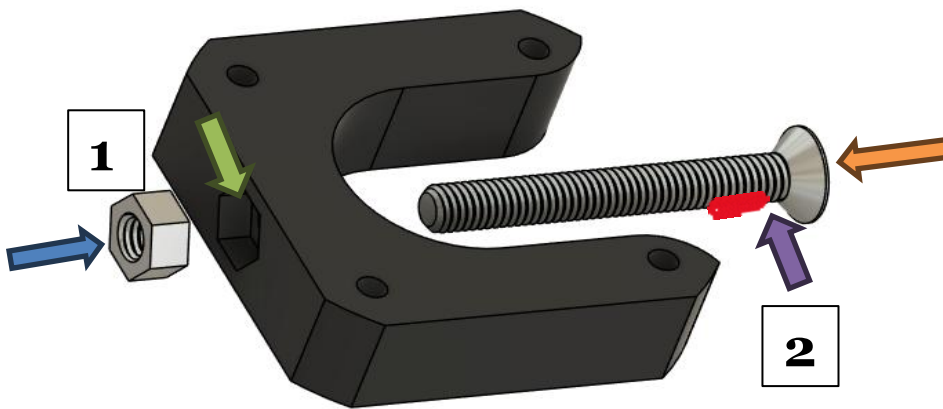
The rear arm is a bit different, one of the rod will be shorter

This step is not mandatory, tests has been made and even without those rods, no issues on the arms happened even with very heavy loads.

At this point you can install the wipers, and the top retainer, and the belt grabber and wire organizer following the Vcore Rat rig manual since nothing changed

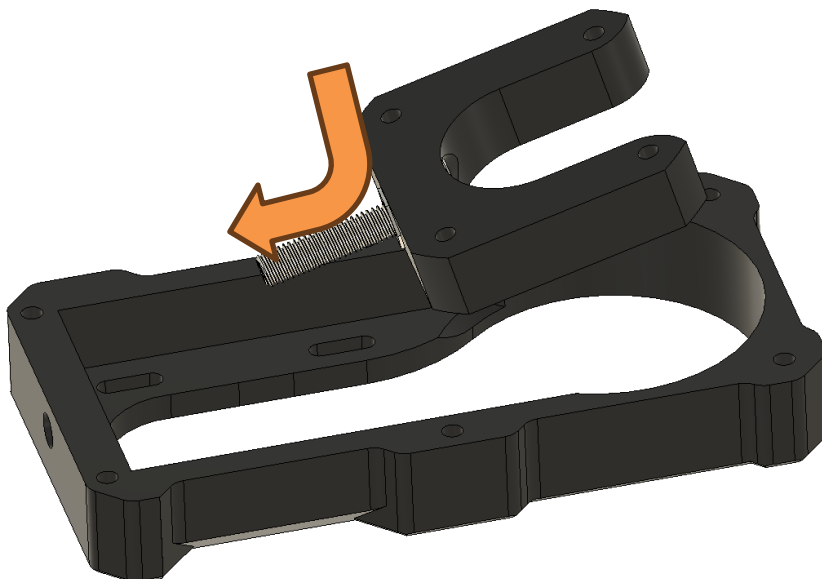
3 Tensionner + Bottom cage

Note: The HD version and legacy version are using the same design but with different thickness here! Same logic anyway

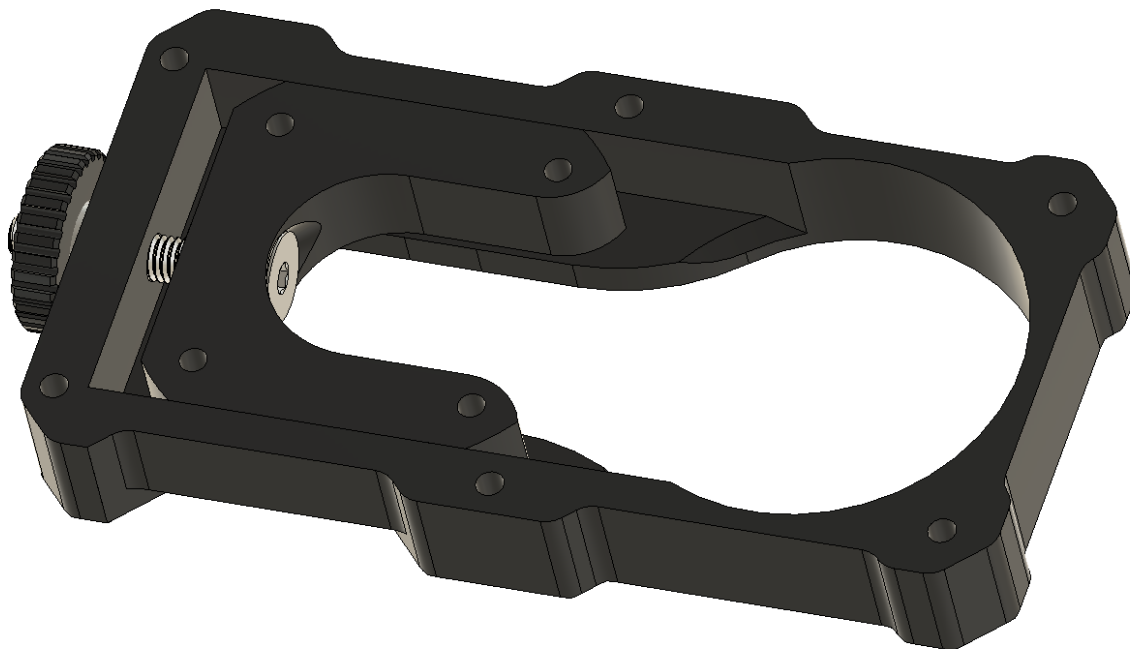


Apply glue in the nut part location. Install a M5 nut or a Nyloc M5, then let it dry 5 min.
Apply glue or threadlocker on the first 6mm of the screw. Install the screw fully! Let it dry 10min

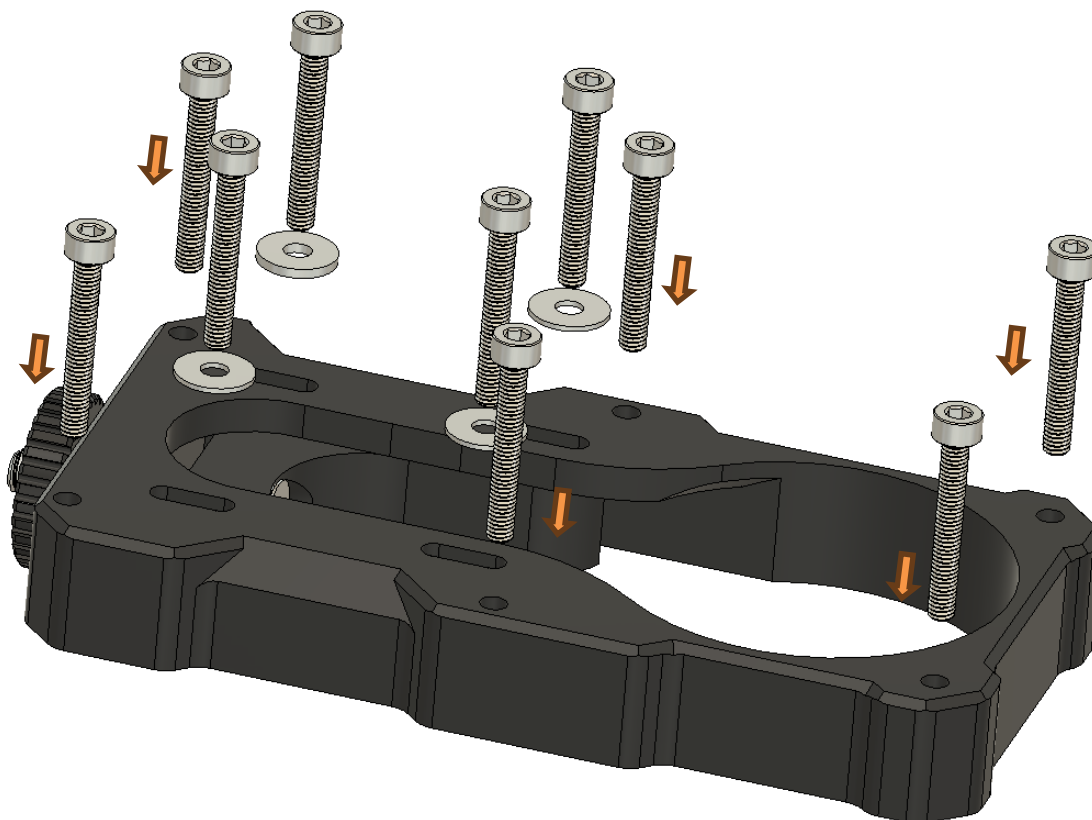
Place in inside the bottom cage

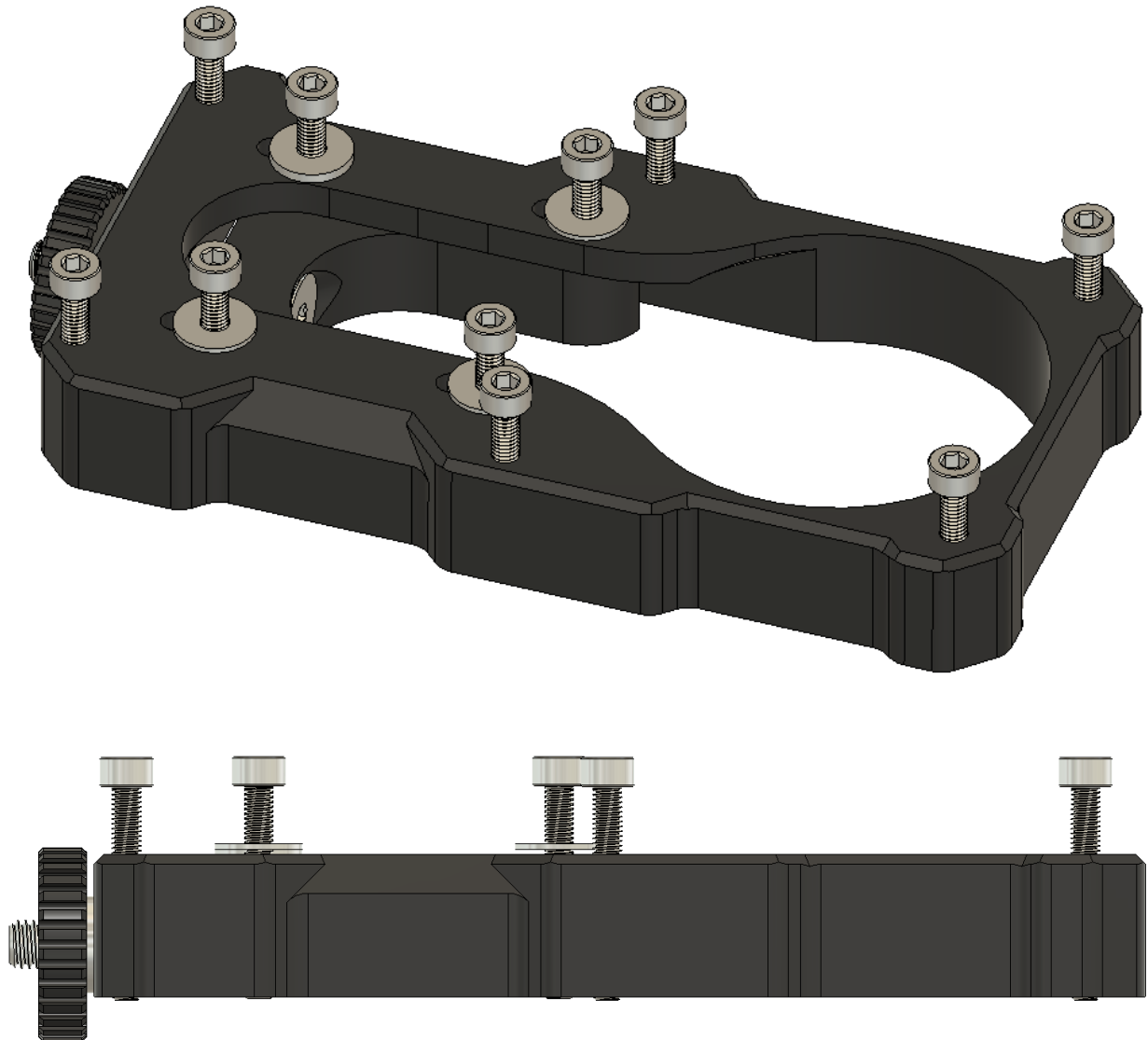


Add the washer and the knob, but don't tight it yet



Preinstall the screws for later



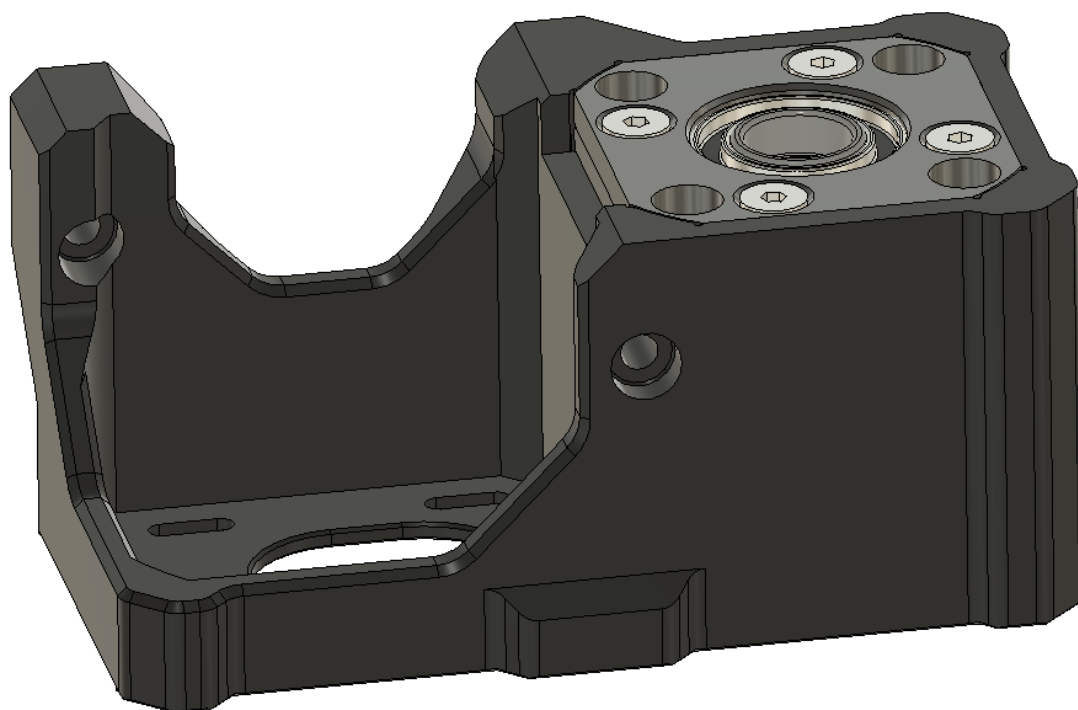
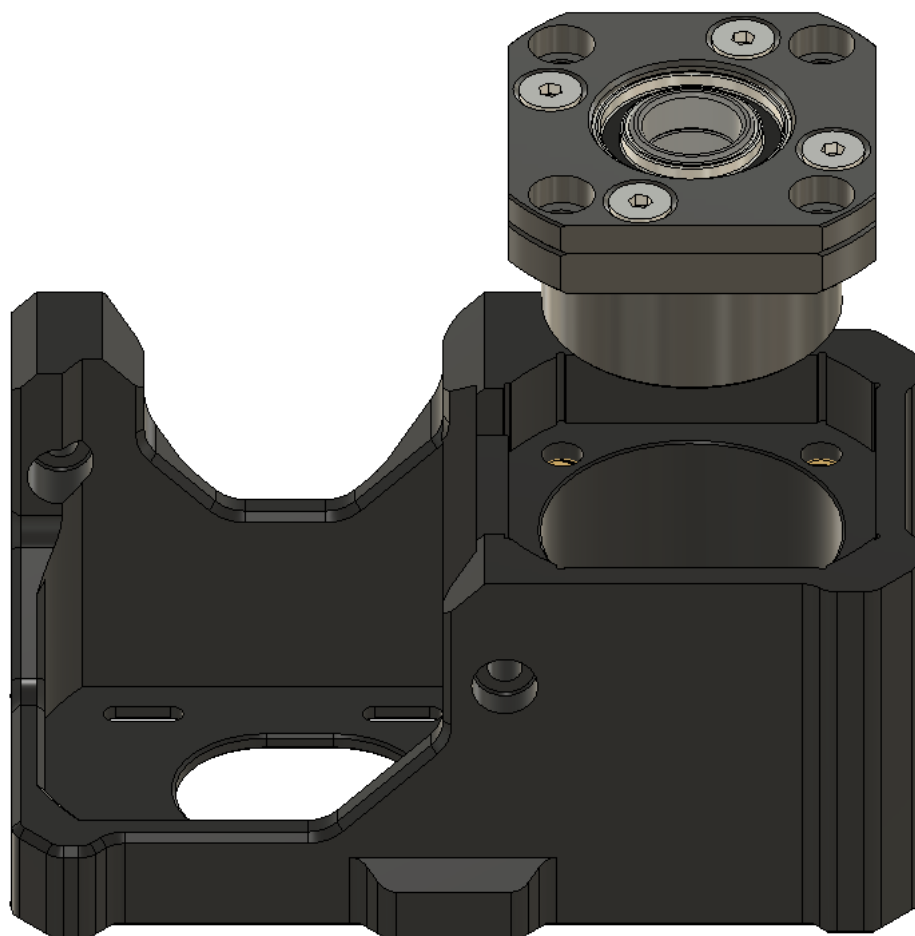


4 Blocks installation

Note: The easier way to do it is to lean the machine on the back or the side, but that is completely up to you depending the status of the machine (Kit or already assembled)

A/ 1605 version:

1605 version features the standard full metal FK12 to place in the block:



B/ 1204 version:

Here we need to use a small variant; as hole would not self align, you need to make the FK10 replacement with the printed version, featuring correct holes alignment

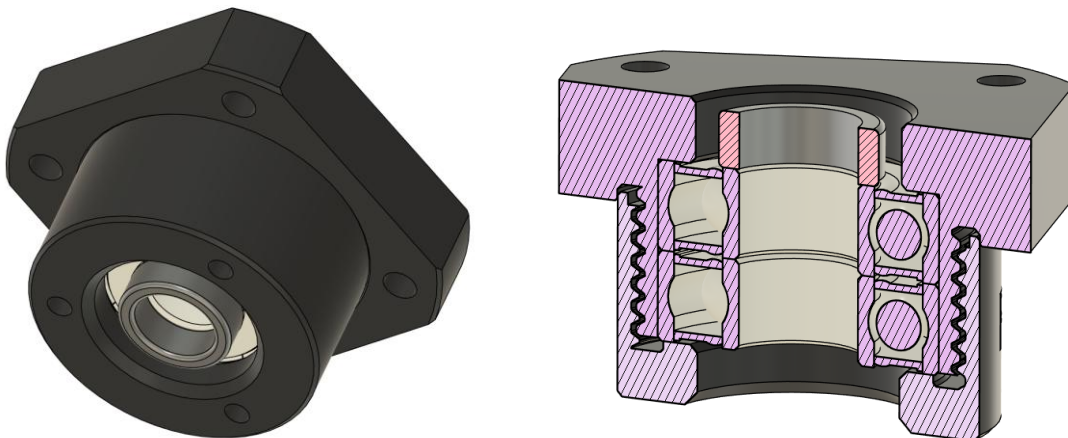
The FK10 custom is made with 2 parts, which are screwing on themselves to completely secure the whole as a block

This way the bearing are completely constrained, and no hardware is needed



Pressfit the 2 6000zz (or 2RS) bearing (*For DIYer, angular contact or standard can be used*)

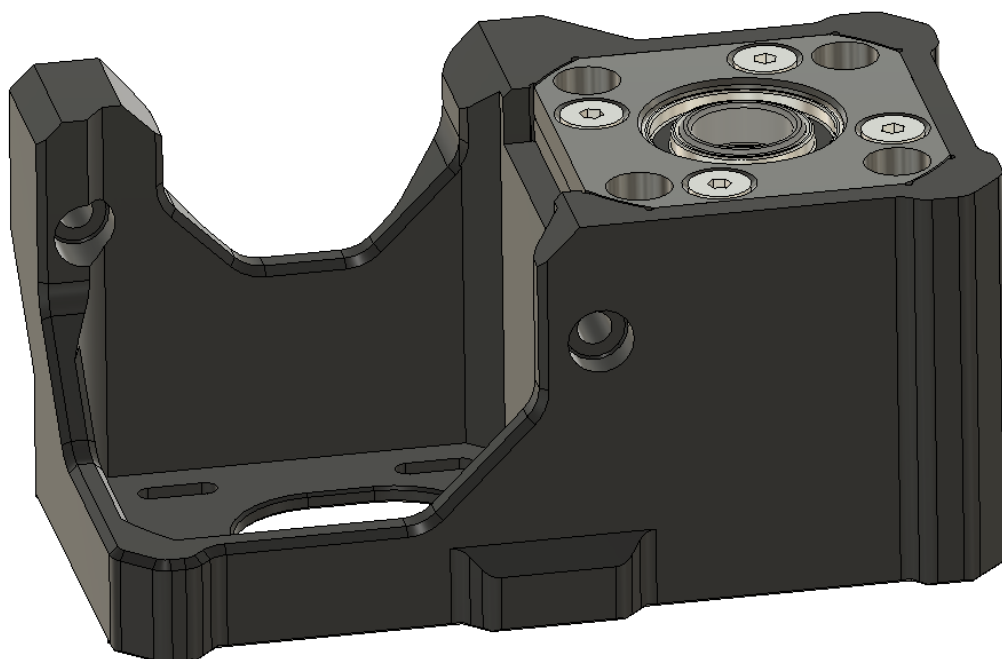
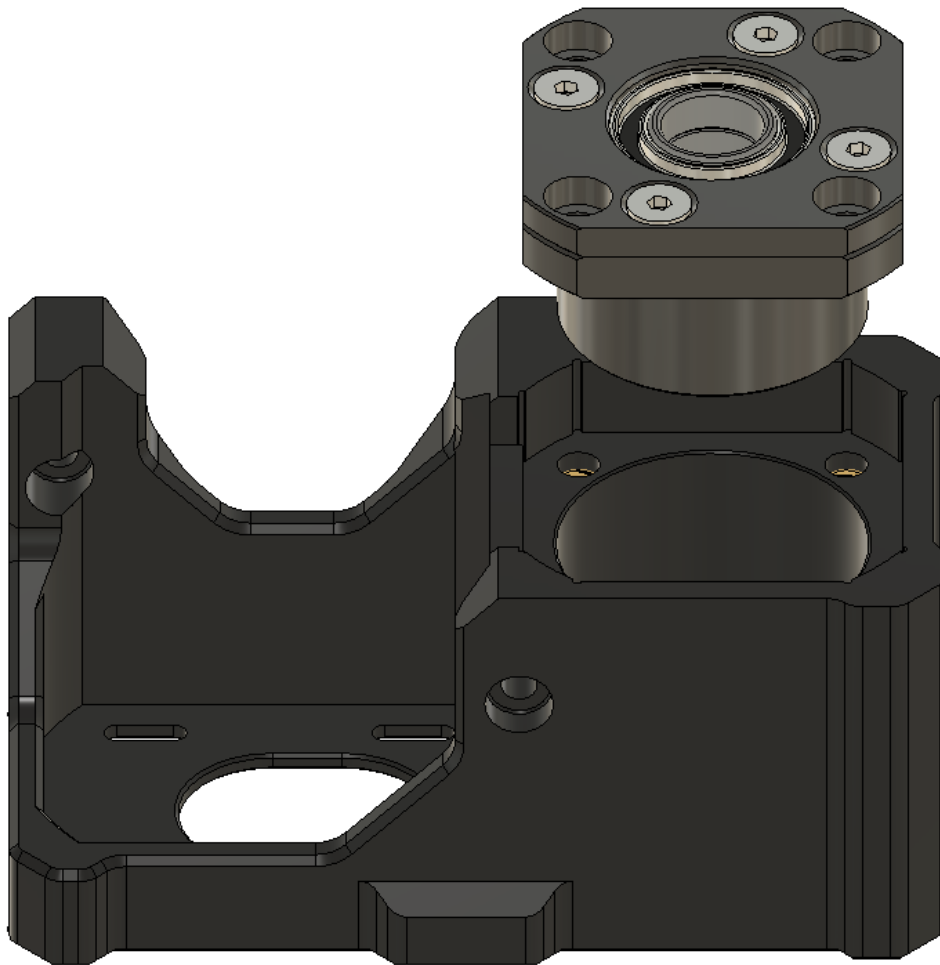
Don't install the spacer under, as the locknut afterwards, since the last revision we don't use them anymore



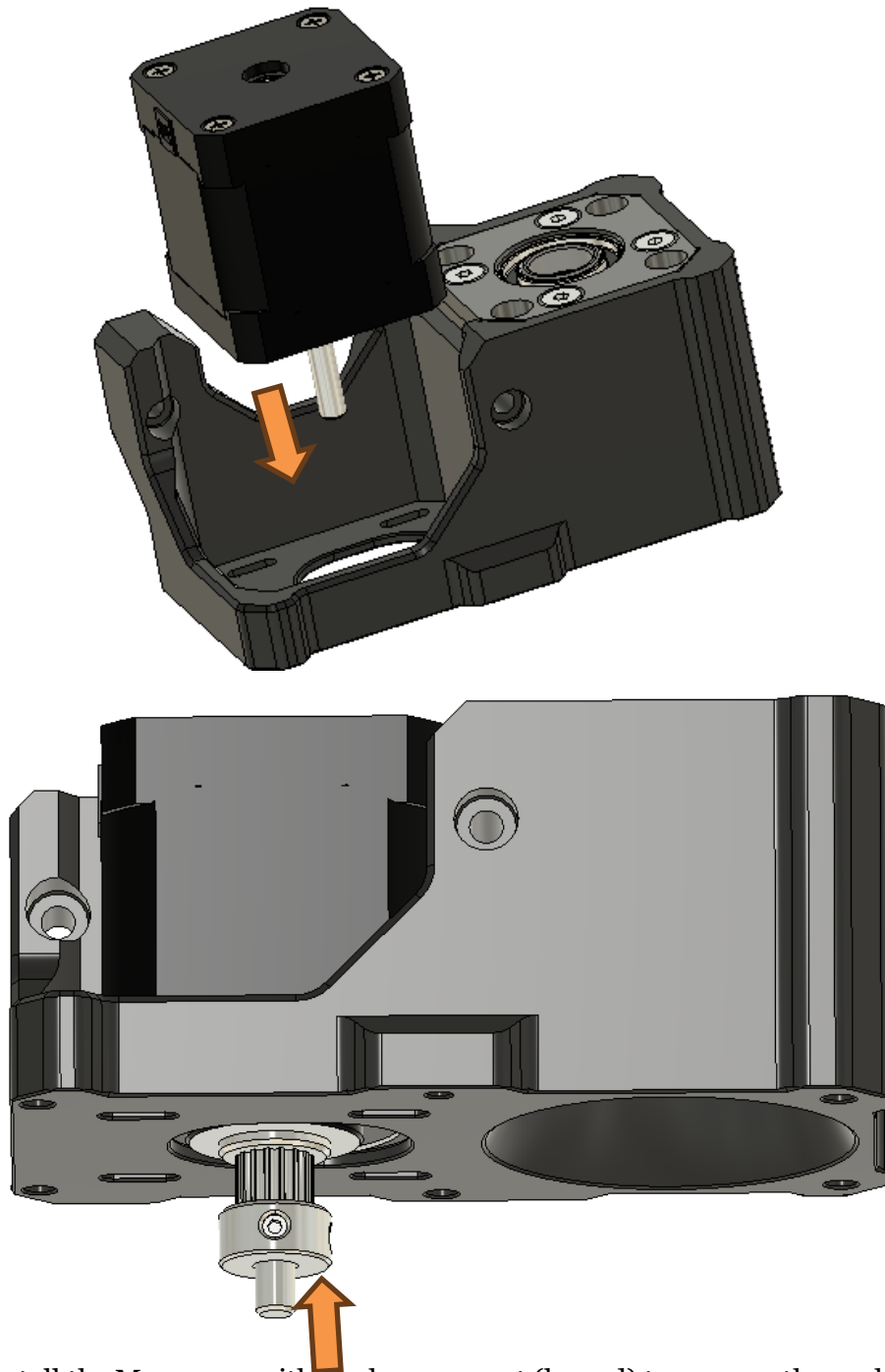


Use this tool (printed with 3 3mm pins) to tight the bottom section clockwise and secure it

Then you can install it the same way than the 1605 variant

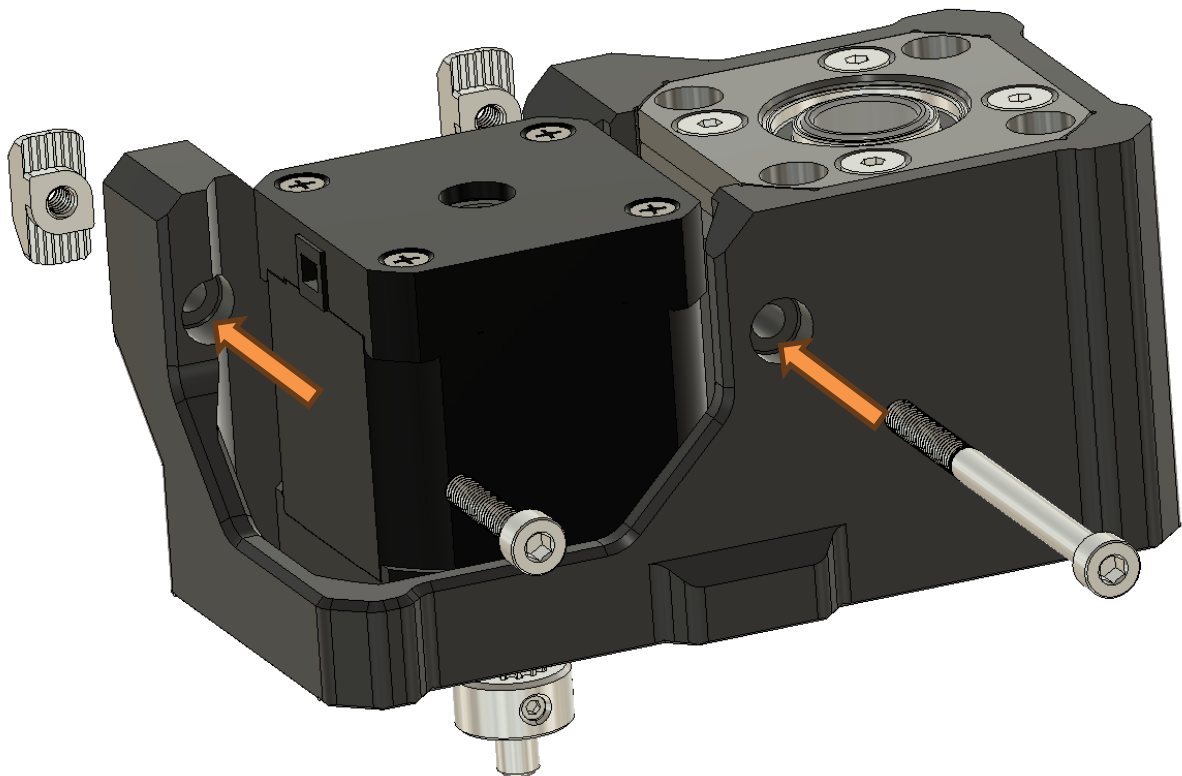


Install the NEMA17 (remember this model accept up to a 48mm motor, not more), and attach the 20T pulley (0.5mm from the motor base).

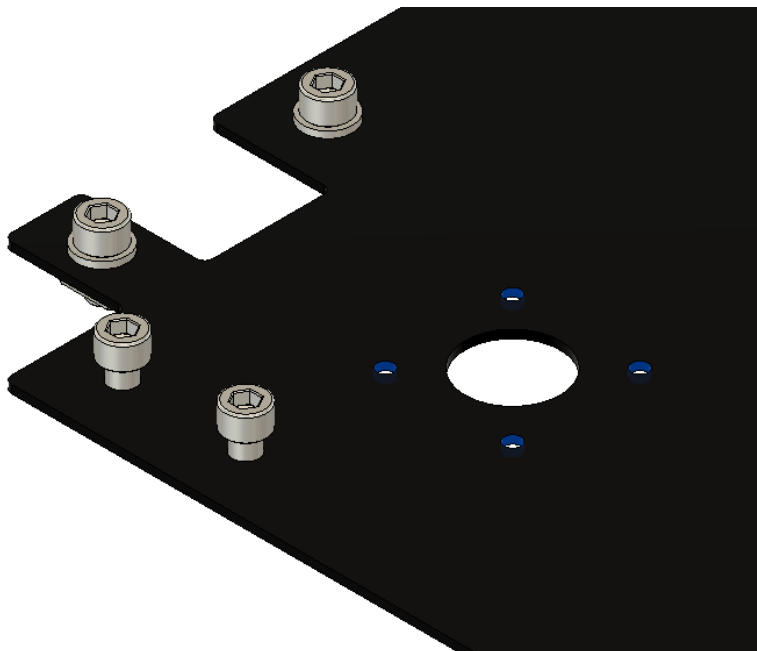


Install the M4 screws with the hammer nut (loosed) to prepare the anchoring

M4x60mm and M4x20mm for ech front arm, M4x20 and M4x90 for the rear block

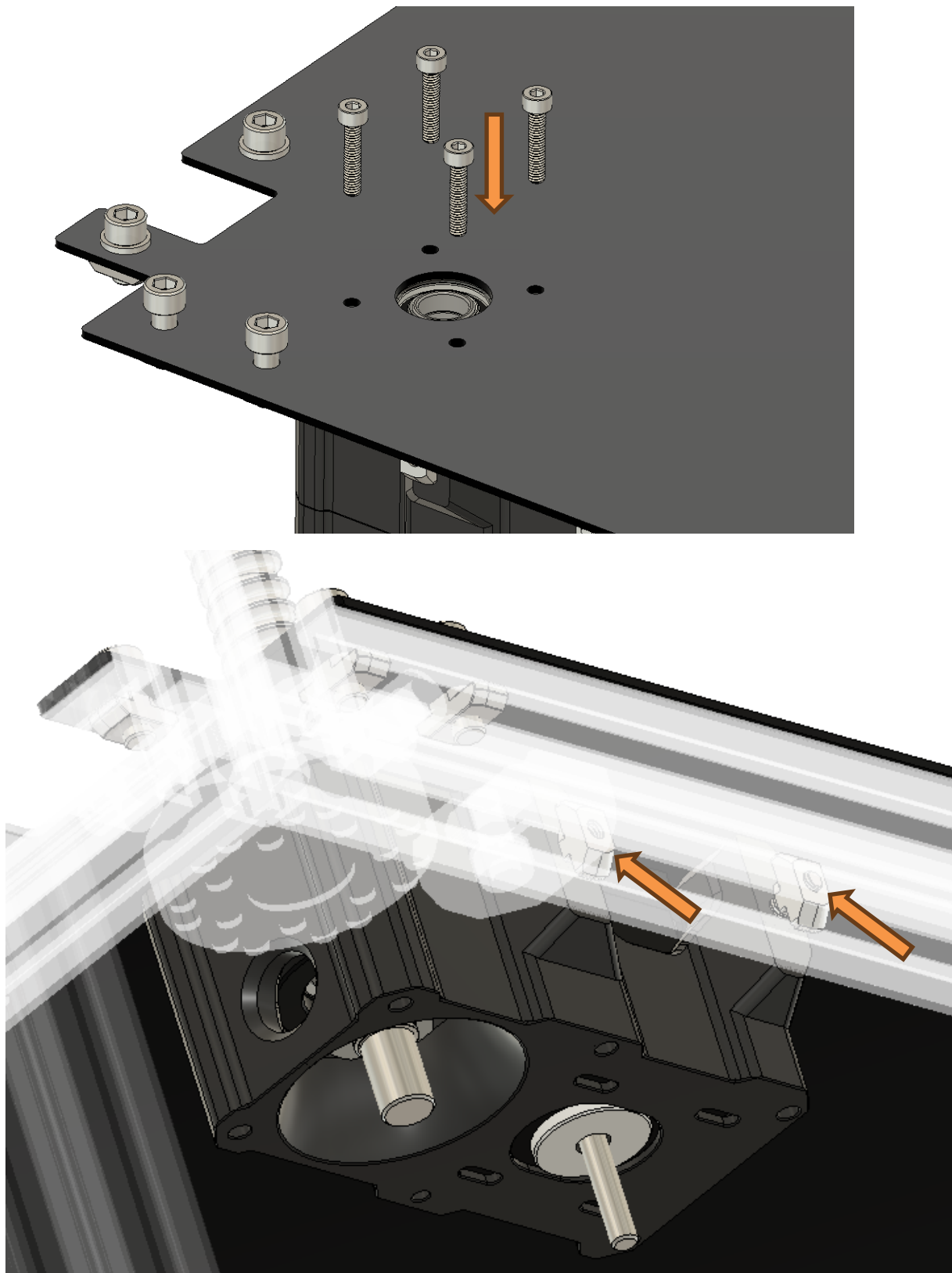


Enlarge the 3mm holes to 4mm with a 4mm drill bit



You can now place the main block in position, secure it with the 4x M4x20 screw through the plate, and the lateral screws in the extrusions with the hammer nuts

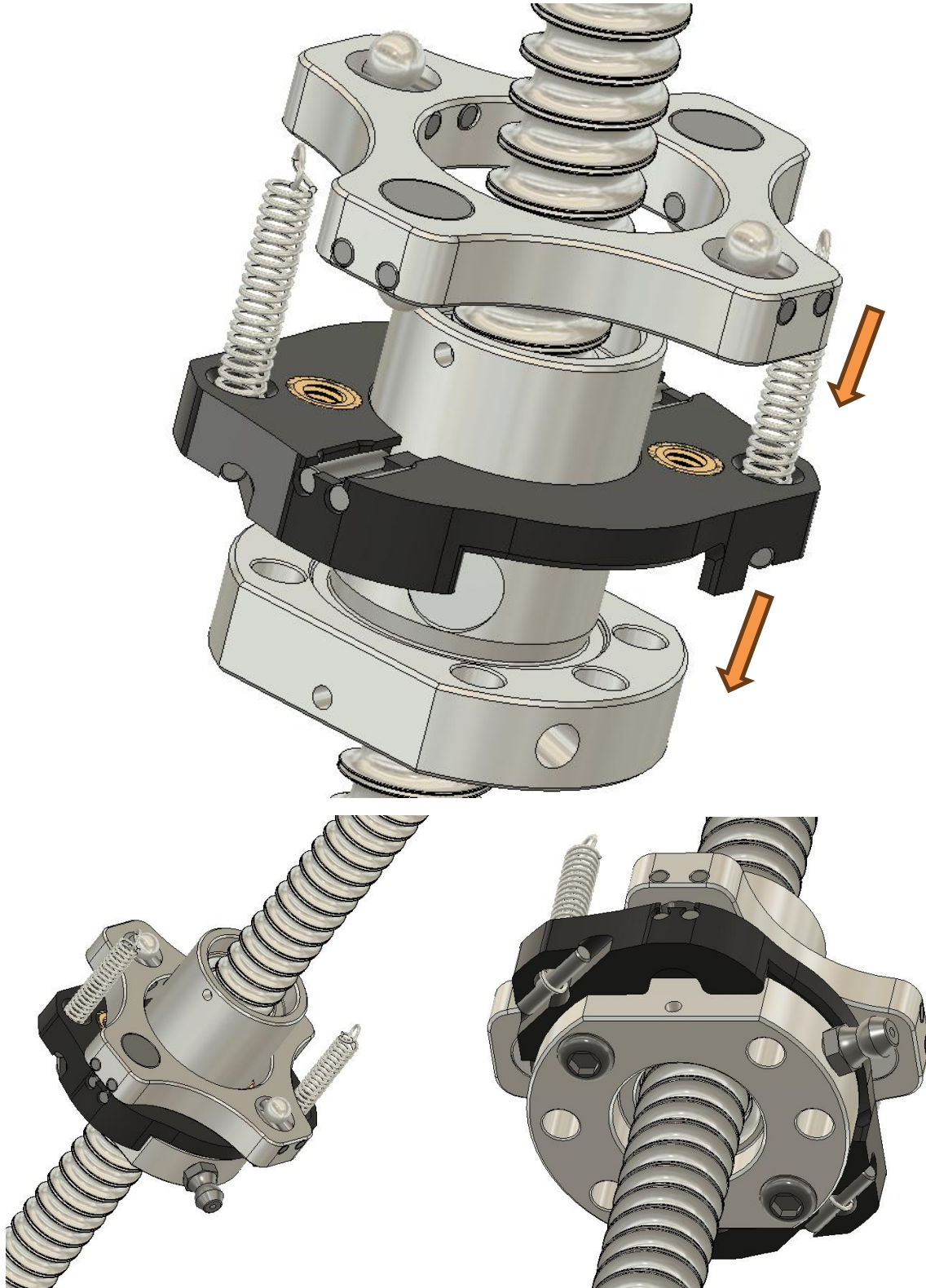
NOTE: M4x40 are to be used with the HD version here



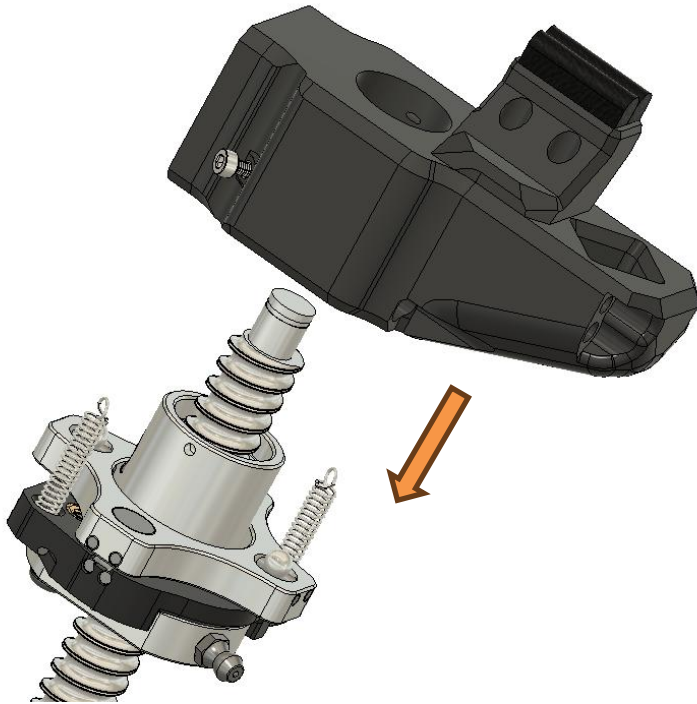
Repeat this process for the 3 blocks

5 Arms installation

On each ballscrew, installt the low ring and the WobbleX



Take the spindle assembly and insert inside the arms hole



Details



Pins locations

For the wobbleX (12/16), same logic: Add some M5x20 to fix the BS nut to the lower ring, add the WobbleX middle ring and install it the same way with the spindle motion seen above.

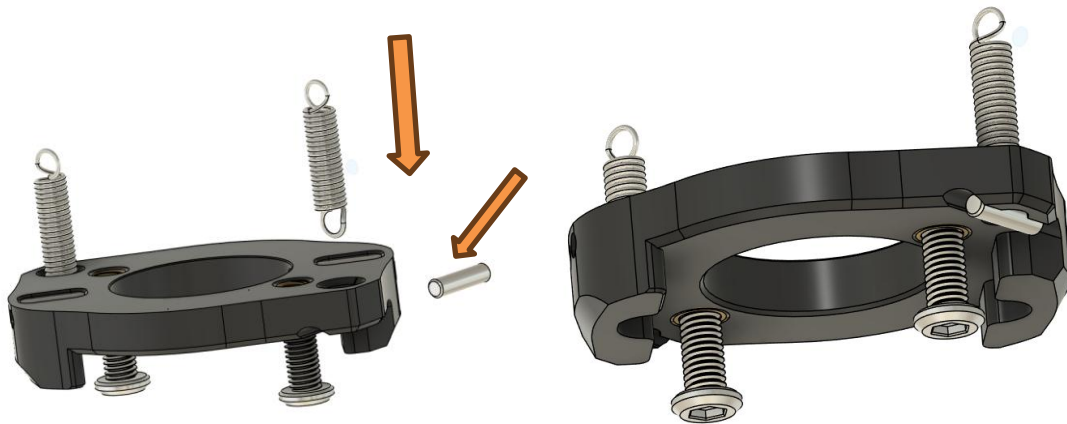
Be careful: 1204 and 1605 version got some differences:

-Discs are larger, and use a different layout, 1605 and 1204 are not compatible between each others, M5x16 are used to secure the ring, with 1204 you need the m4x16

-Arms got a different hole layout and dimension as their pins position

At this point you can change the methods, as few exist to get to the same objective.

3.2D revision :

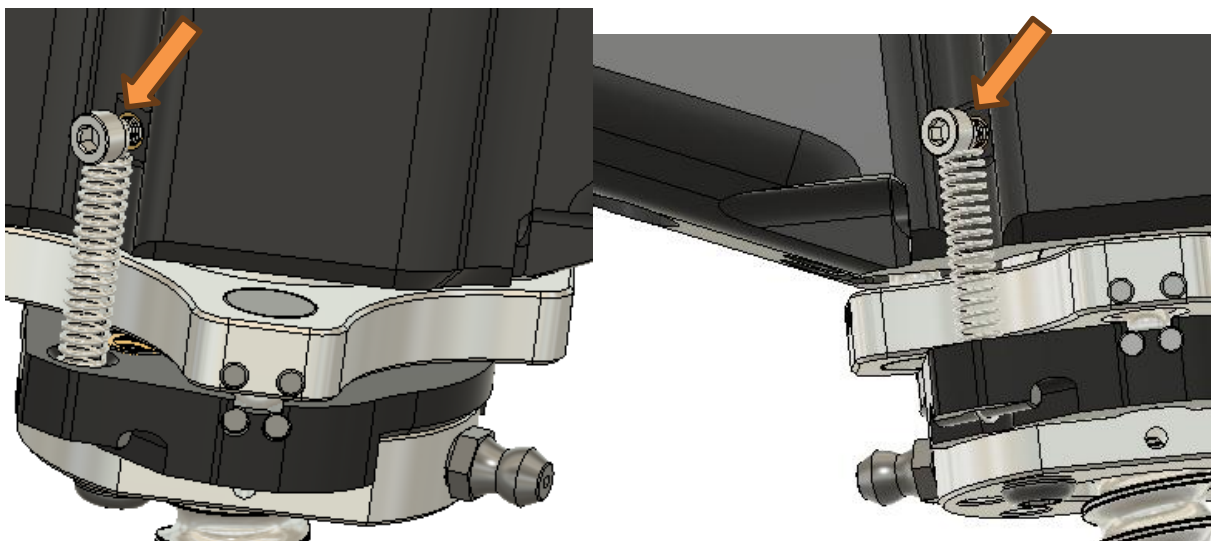


The last revision features a spring retainer to avoid lifting the full assembly and loose the ball from the WobbleX

Insert the spring in the hole, secure it with a pin

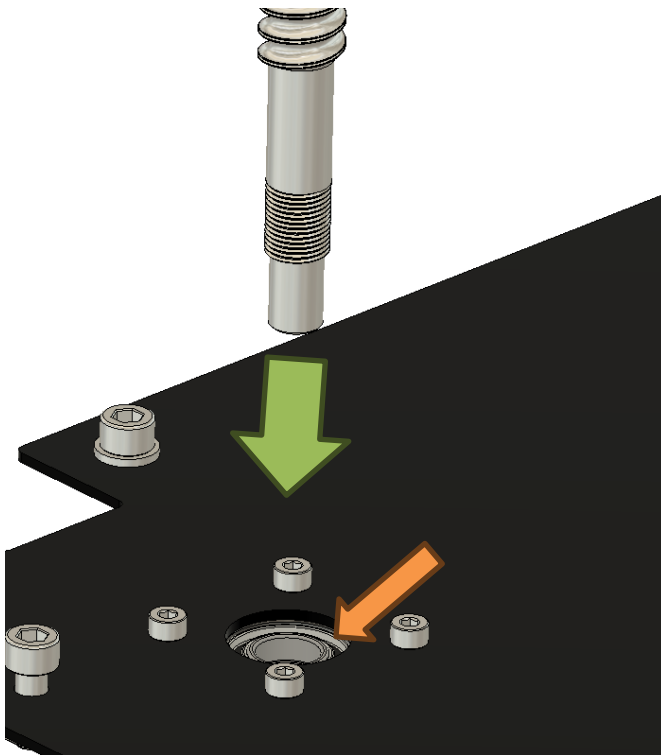
This under ring feature only 2 M5 screw new, proved to be enough in this case.

At the end of the unit assembly, use a plier to tension the spring and reach the arm mount point



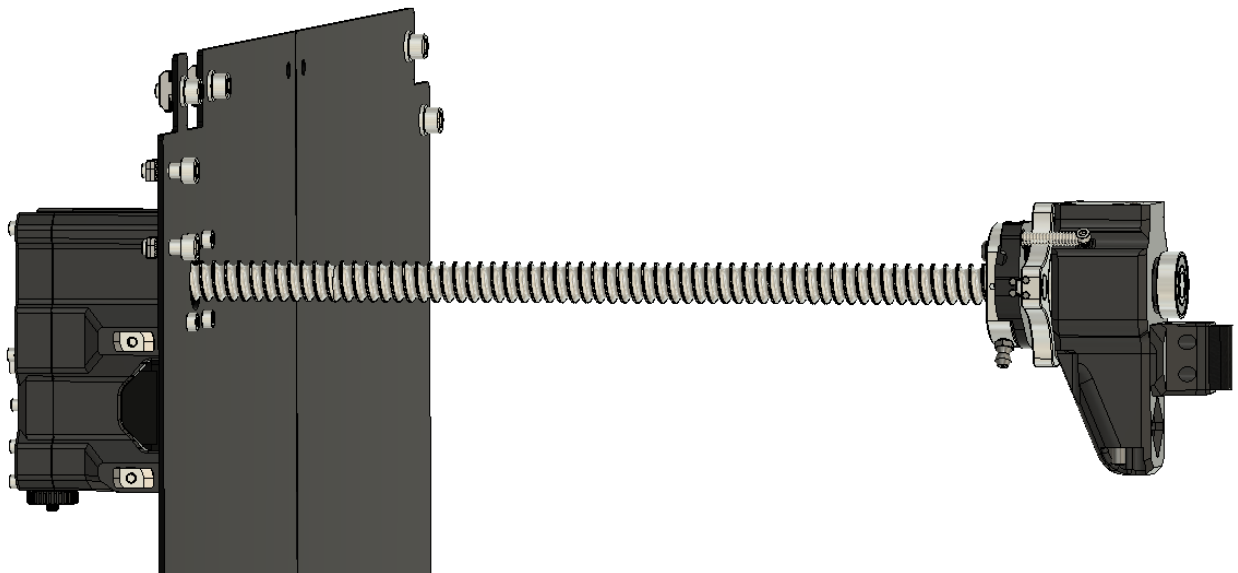
6 Ballscrews installation

Now insert the other machined end inside the plate hole, until the ballcrew get fixed in position

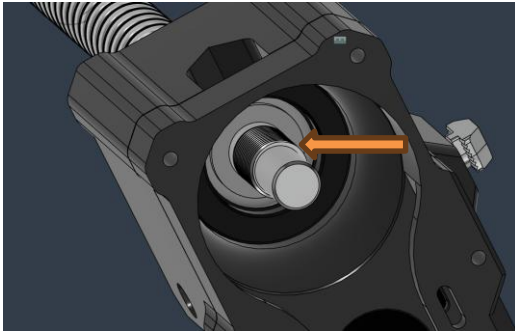


This should look like that

The 1204 custom FK10 will need the **5mm spacers** on top of the 6000s bearings, don't forget them.



Secure it with the locknut bellow the block



On the last revision, **DON'T INSTALL THE LOCK NUT AND THE SPACER**

In some case scenario, the locknut can induce a wobble like artifact on Z with the overconstrain applied to the spindle

As the deletion of it solve the issue, and since this locknut don't bring anything to the design, choice have been made to not use it

Same thing for the lower 5mm space, which is no longer needed

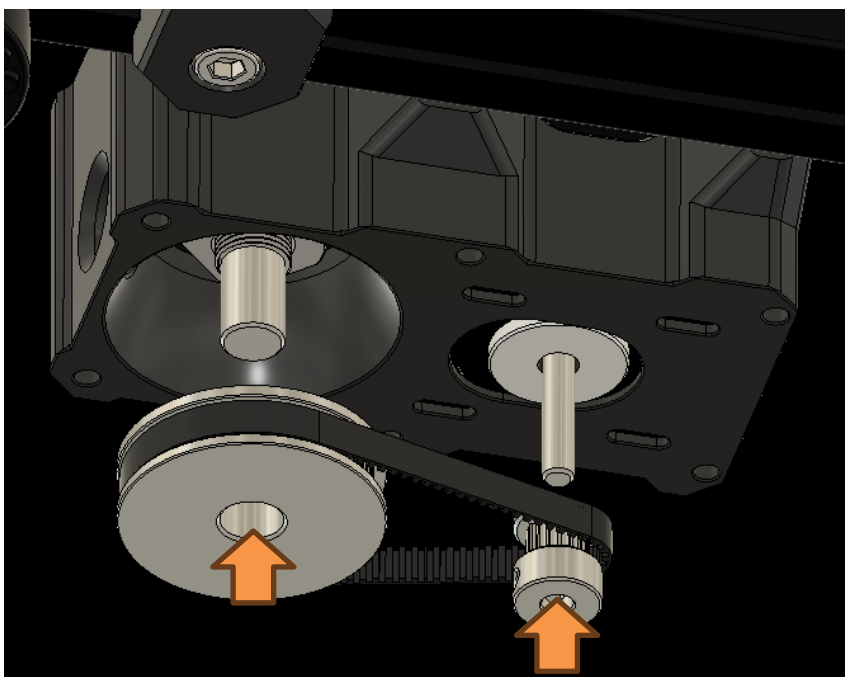
Locknut doesn't have any importance here since we stand vertical, no thrust will appear so the total bed + spindle weight will do the job, and remove a potential source of issues

7 Drive closing

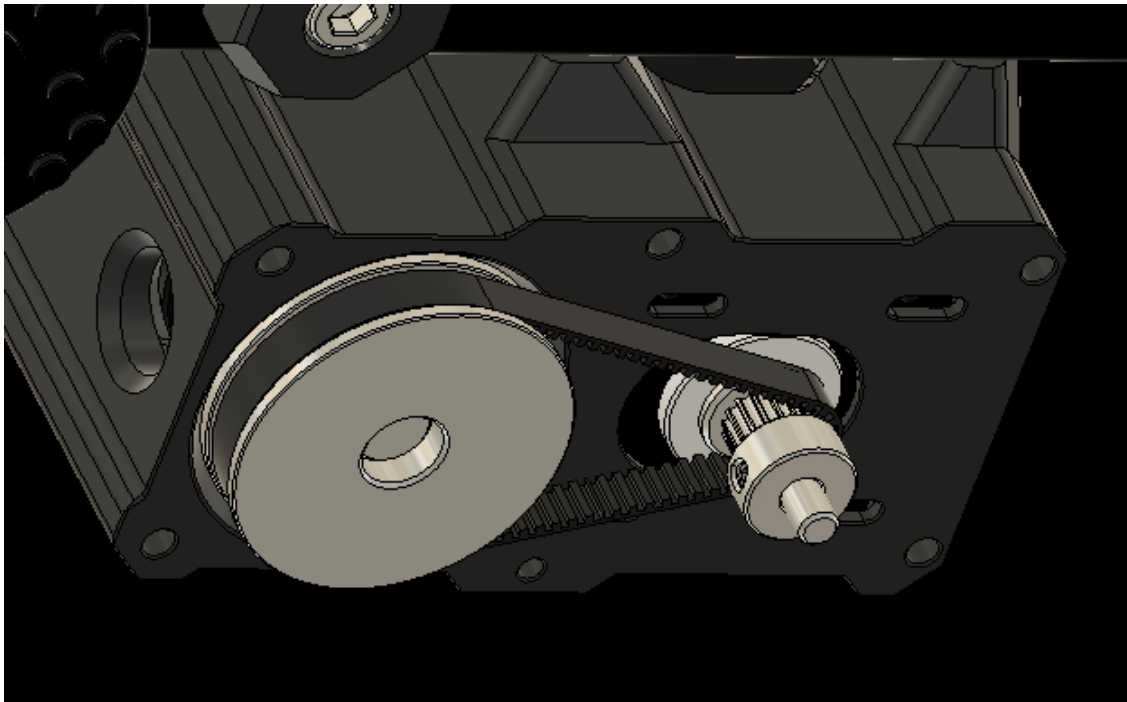
At this point we can take care of the drive system

Install the pulley with the belt (8mm bore for 1204/ 10mm bore for 1605)

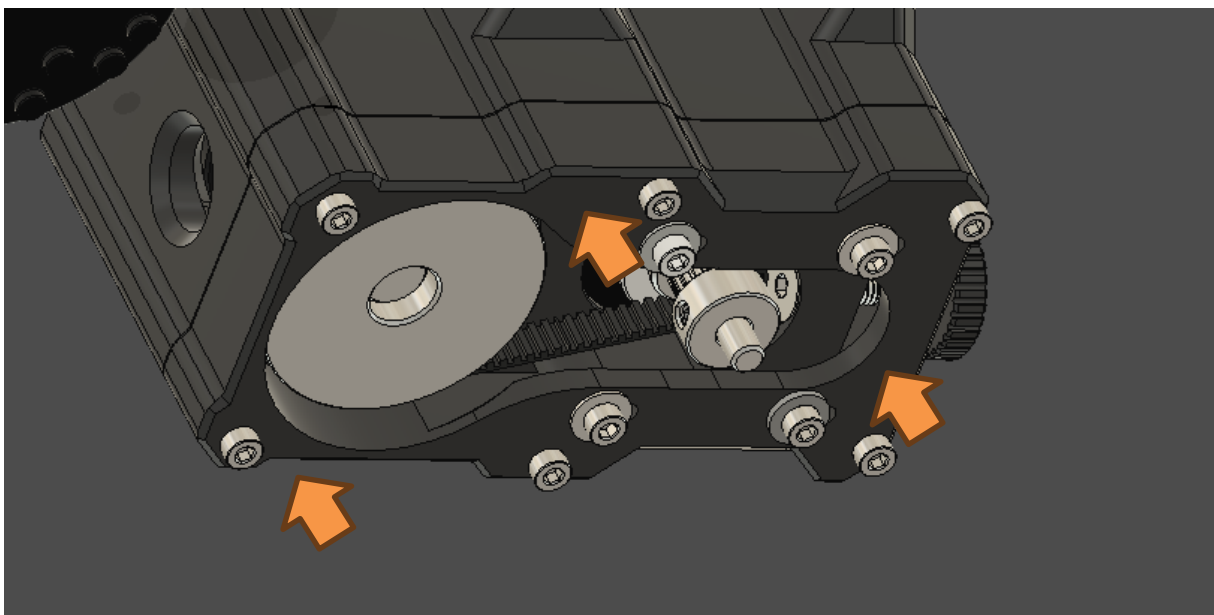
BE CAREFUL WITH THE 60T INSERTION, DO NOT FORCE IT, ALUMINIUM CAN BE DEFORMED BY THE STEEL SPINDLE



Position ends like that



Then install the bottom assembly made earlier



This process need to be done on each 3 Z axis

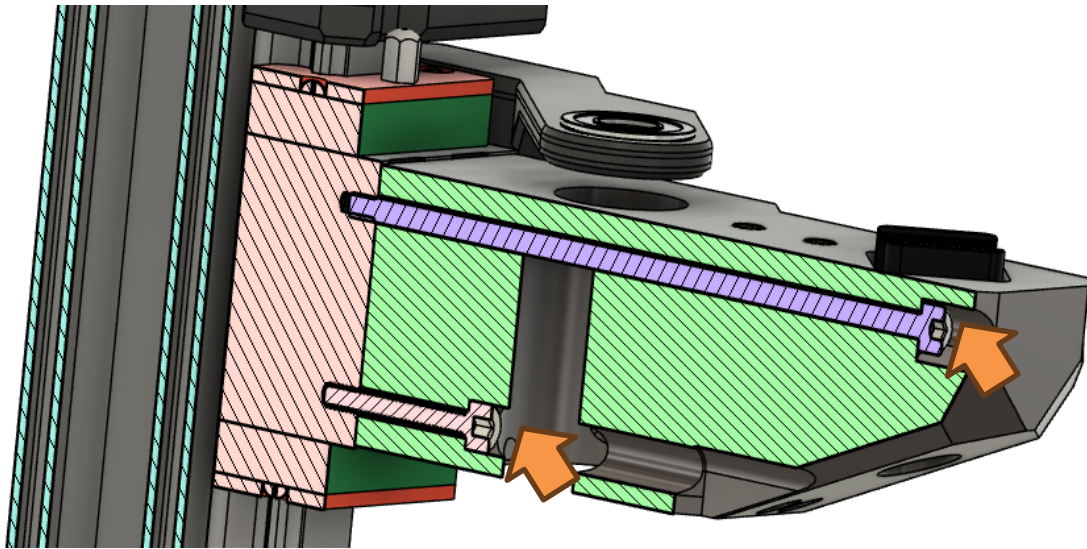
Note; don't overtighten this cage, it is maily a protective cover and guide for the tensionner

8 Arms to the HG15 carriages

For the arms, same logic than a regular VC4 kit

Use the long screw from the Ratrig kit for the 2x top retaining holes

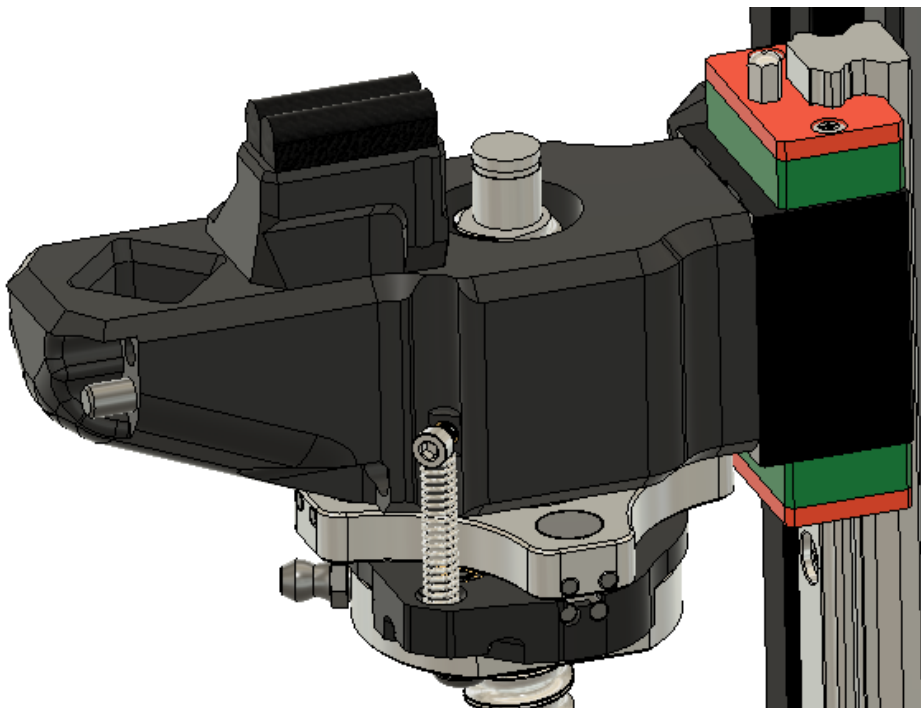
And use 2 M4x20 on the bottom retaining hole



Exemple figure

Once installed, you can hook the springs end on the m3 screw from the arm an get the sytem ready

You should end like that

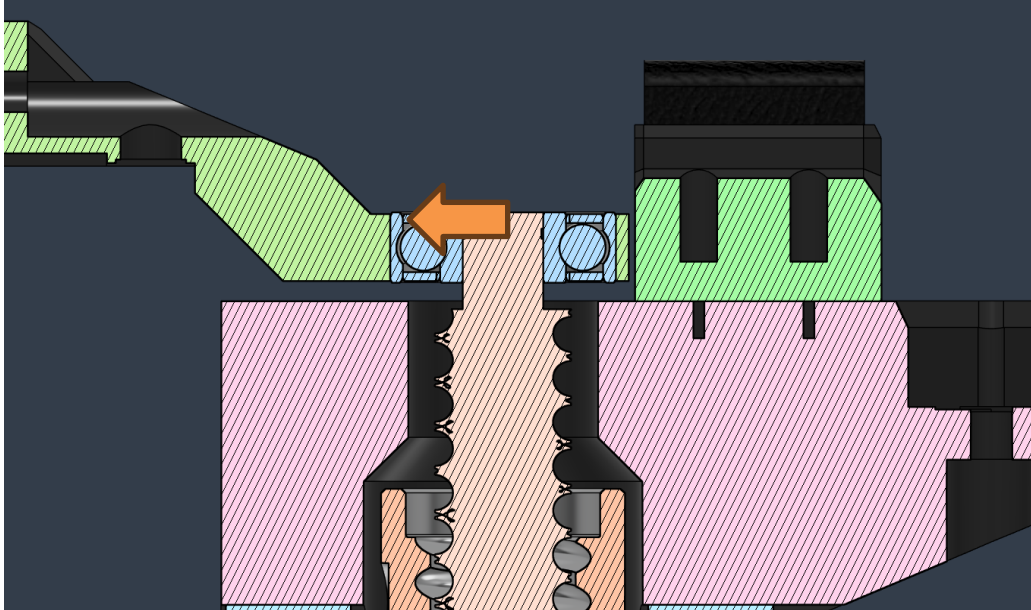


9 Top Z constrain

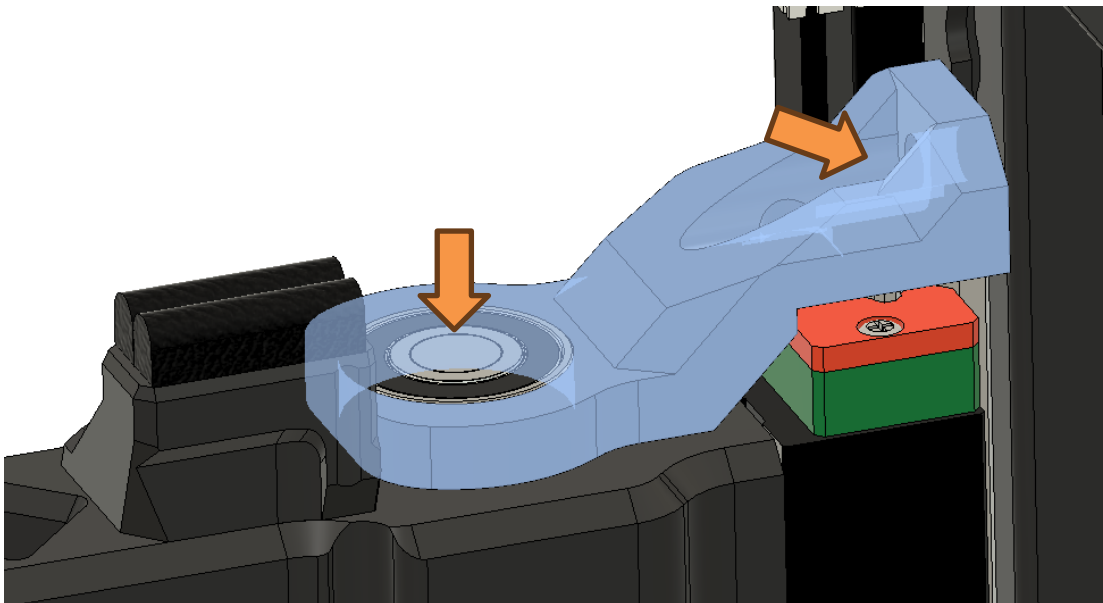
Insert the 6000 bearing for the 1605 variant

/ 608 for the 1204 variant

A gap can exists between the bearing and the end seat, we want it to be here since it will compensate thermal expansion on heated chanber and avoid Z bending in some case scenario



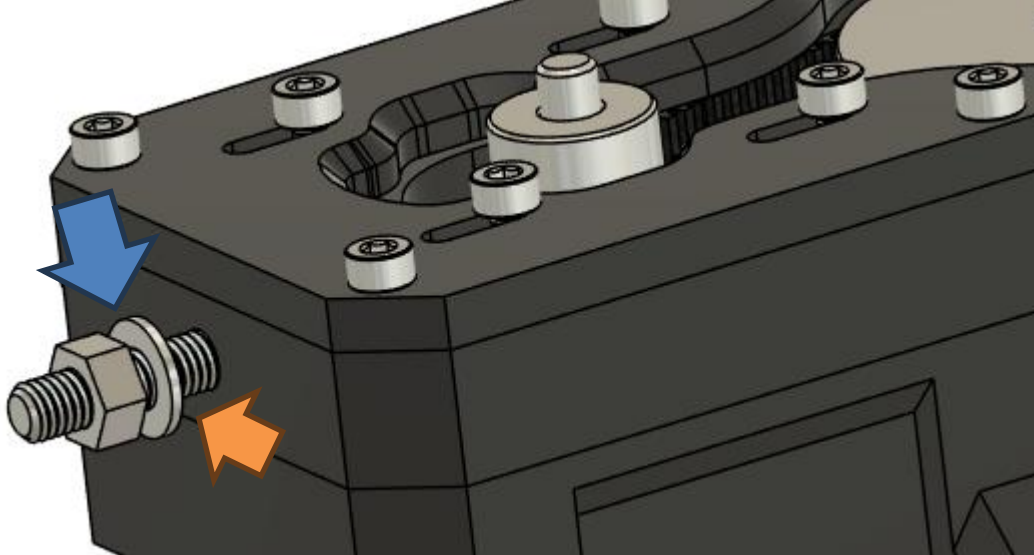
Same method that the Ratrig manual for the printed part



Repeat that on a Z axis, same parts same method

10 Tension the Z drive

- 11 Push the M5 bolt to gain the minimal position.



Now tune the belt tension through the M5 Nut + washer here (Use the knob if you prefer)

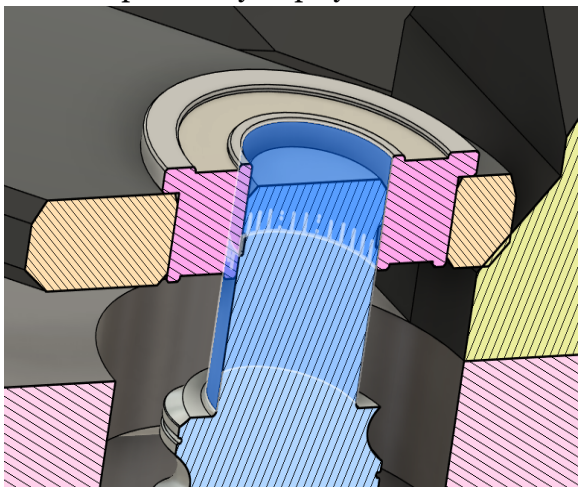
We need it to be tight but do not overtension it either, we want a firm flex on the belt.

Repeat the process with the 2 other remaining block

12 Final assembly

For the 1204 version, simply keep the standard stock LS/BS retainer at the top, we use the same ID 8mm bearing

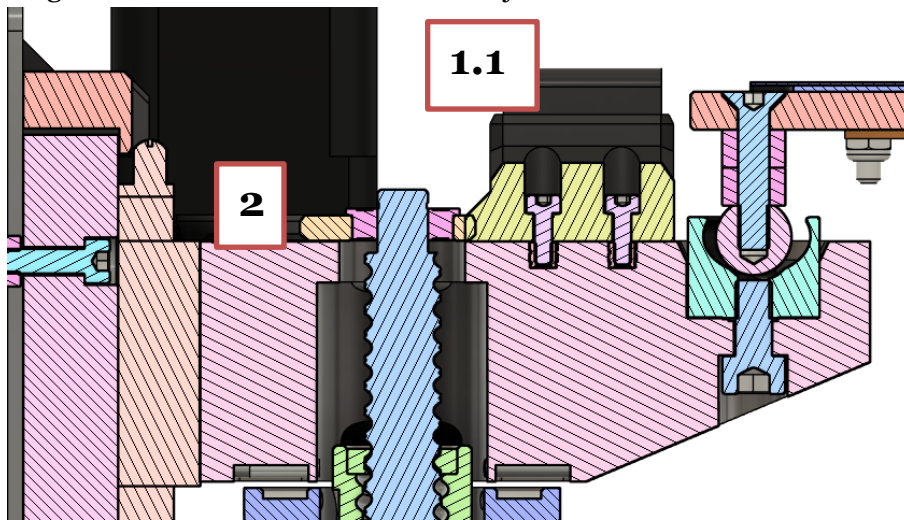
In order to get some safety for potential collision, place the retainer with 2mm margin, you have the possibility to play with the retainer position



You can adjust a bit the height of them if necessary.

You can at this point add the bearing balls into the WobbleX assembly if not done before

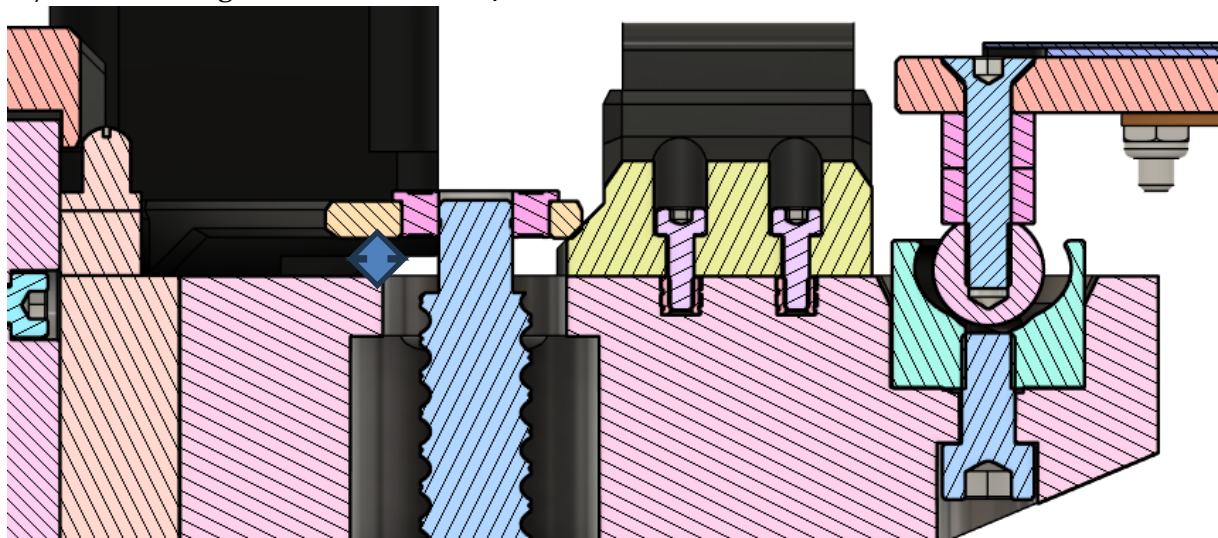
-Depending the Hot end you will use (Rapido, Magnum, Goliath, ...) or a custom bed, the max height can be different. There some adjustments that can be done to maximize it



-1/ Add some M5 washers on the bed feet, to add the necessary height missing

-1.1/ Elevate the wiper if necessary

-2/ Tune the height the the retainer: 4mm are available



13 Config cfg

The first thing to do is to correct the **rotation_distance** of all Z sections

1204: **4, transformed with the 1/3 ration to +1.3333**

1605: **5, transformed with the 1/3 ration to +1.6666**

Then we need to reverse the Z motors since they are inverted (pointing the bottom)

Simply add a “!” in front of the “**dir_pin**” (ex: **dir_pin: !PC12**)

Notes:

-In some rare scenario lowering the RUN_CURRENT on the Z motors, can improve the behaving of the system

-Remember that we do not need power here since the 60:30 reduction is working for it already

-I strongly recommend placing safe, low temp focuses settings on the TMC section of the Z motors to get perfect

-Use 16 microstep

-Keep in mind the NEMA + the 60:20 reduction + the 4-5mm per rotation motion is decupling the power of the system and can move powerful forces.

-Be careful with your fingers, your toolhead and you light gantry

-In order to avoid overdamages, the M4 insert in the block holding the whole assembly will popout to avoid issues on expensive parts like your gantry or toolhead

14 Troubleshooting

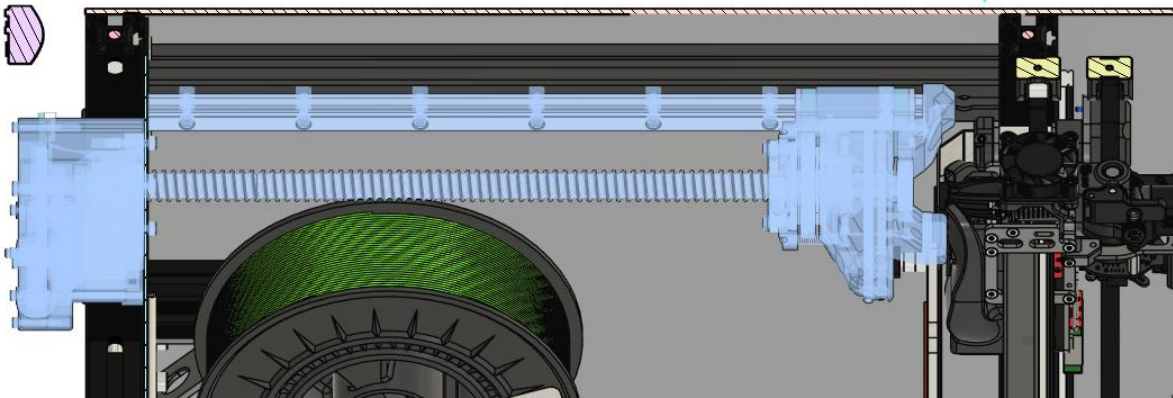
If you encounter some wobble issues:

The Z-Upgrade is made to LIMIT the wobbling, that mean virtually there will be some, at a certain point it is not quantifiable

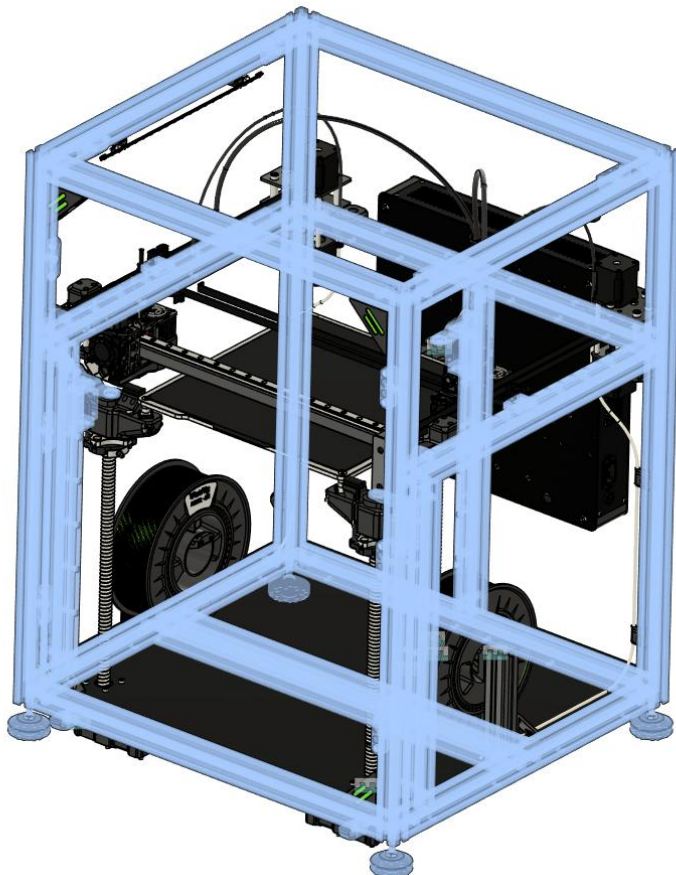
The level of Z artifacts is link to the quality of the assembly

Actually the upgrade was release after a lot of test, and a lot of beta testers and tremendous unit since 2021 behave perfectly. So it works when everything is lined up accordingly

- o- The alignment of the lower bracket and the retainer has to be perfect
Check the alignments, in particular the RAIL / Screws
VCORE 4 and 4.1 with the ZU4 has been made very hard to miss the alignment, but still don't forget to check

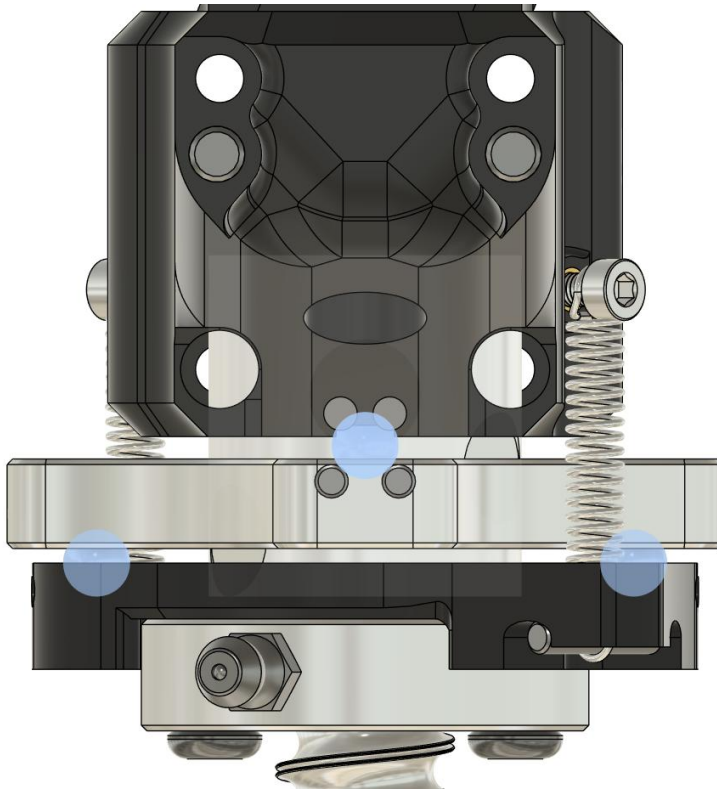


- 1- The squariness of the frame can induce Z light wobbling ; try to square the Frame the better you can

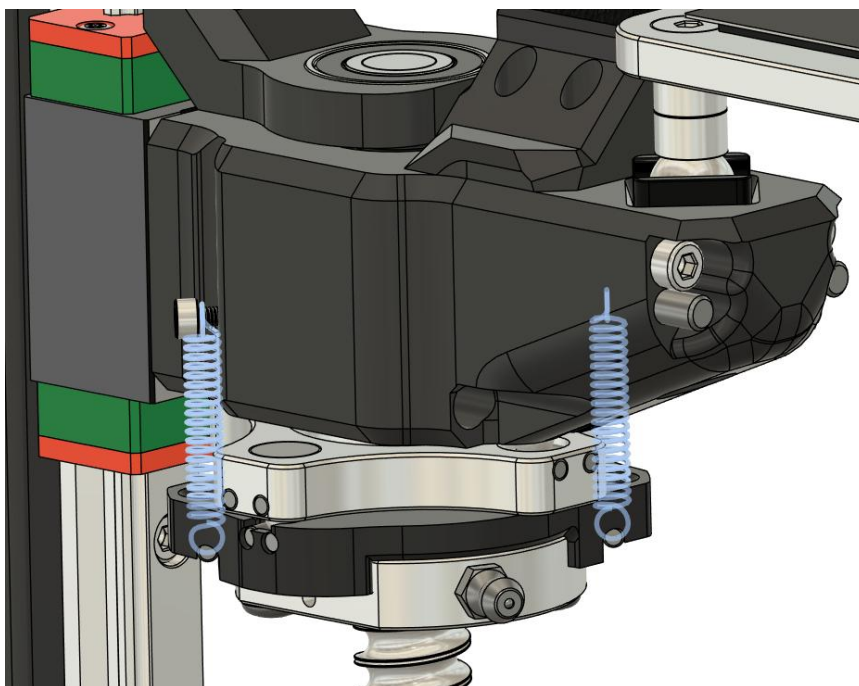


- 2- Check if there is no collision to the wobble system and the SFU Nut in the free space arounds it

Check the ball seats



- 3- Remove the springs; As they slightly increasing friction, it can make another issue worse, It will be working perfectly fine without, as they were meant to secure the ball seats in the system



	Z-Upgrade 4.0 for VCore 4.0/4.1	BRS-Engineering

- 3- **Check the tightening** of screws, nema etc
- 4- Hard point checks, the spindle can be rotate by hand without resistance
- 5- Cleanliness / lubrication checks (Rails + BS): Use a Hiwin GSo4 lithium type grease or qualitative equivalent

Disclaimer

The system is designed to operate on a properly assembled Vcore. Even a slight mounting error can make it impossible to upgrade or issue during prints.

If the parts to be printed are made by the customer, check that the dimensions at the output of the printer are respected. : a bad dimension will block (+ -0.25mm) the assembly.

This kit is an optional upgrade, intended for an informed public and having advanced experience, its assembly and / or its function and / or its quality of execution are the responsibility of the customer and are not guaranteed in public view by BRS-E. BRS Engineering accepts no responsibility in the event of bad sourcing (bad quality and / or bad dimensions sfu), bad assembly by the customer, or bad assembly of the base Vcore.

The kit has proven its POC and POW in quality controls at BRS-E as well as at a beta test program, As is, the design works as expected

15 Assistance

If you need assistance ; I can help on:

Mail : contact@brs-engineering.com,

Discord: BRS-ENGINEERING-Florent Broise
brsengineeringflorentbroise_3873

Meta Messenger: BRS-Engineering